Report

Wilfred Allison

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Guiding Questions

Question 1: Linear regression analysis makes several assumptions. For one, all observations in the data must be independent of each other (e.g., the data should not include more than one observation on any individual/unit). Furthermore, the data should avoid including extreme values since these will skew the results and create a false sense of relationship in the data. In general, linear regression gives more weight to cases that are far from the average. Can you think of any examples or datasets in which this might pose an issue?

This would be bad for a case like medical cost modeling, where people who are extremely sick would vastly sway the model. Or for economic historical data modeling, things like the Great Depression or a bubble would overly affect the model.

Question 2: In this discussion post, the argument is posed that there is no ethically neutral statistical method, with specific reference to linear regression. What is the basis for this argument? Do you agree or disagree? Why?

I'd say linear regression isn't ethically neutral. Every statistical method includes choices about what to optimize, and linear regression's focus on extreme cases might work in some contexts but be problematic in others. All models will contain some bias. The key issue isn't that linear regression is inherently unethical, but it can be used without considering whether its built-in priorities align with their values. We could choose other methods which treat all cases equally, but instead we often use linear regression by default. It's the data scientist's job to be ethical.

Question 3: Suppose a machine learning researcher at a company learns a model to automate the hiring process. This company sells software based on this model to other companies looking to expedite their hiring. However, it is later discovered that the algorithm heavily favors members of a certain class unfairly.

a) In this situation, who should be to blame for the unfair hiring?

b) On whom does the responsibility fall to check the fairness of automated systems?

a) The company that uses it.

b) Responsibility to check comes at each step: researchers must test for bias during development, companies must audit before selling, and purchasing organizations must validate before deployment.